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Inventor(s)

Ballard; Andrew et al.

Solid oxide fuel cell unit

Abstract

The present invention relates to an improved metal supported solid oxide fuel cell unit, fuel cell stacks, fuel cell stack assemblies, and methods of manufacture.

Inventors:	Ballard; Andrew (Horsham, GB), Domanski; Tomasz (Horsham, GB), Harman; Jon (Horsham, GB), Robertson; Alan (Horsham, GB)
Applicant:	Ceres Intellectual Property Company Limited (Horsham, GB)
Family ID:	59896119
Assignee:	Ceres Intellectual Property Company Limited (Horsham, GB)
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Background/Summary

SCOPE OF THE INVENTION

(1) The present invention relates to an improved metal supported solid oxide fuel cell unit, fuel cell stacks, fuel cell stack assemblies, and methods of manufacture.

BACKGROUND OF THE INVENTION

(2) Teachings of fuel cells, fuel cell stacks, fuel cell stack assemblies, and heat exchanger systems, arrangements and methods are well known to one of ordinary skill in the art, and in particular include WO2/35628, WO3/07582, WO2004/089848, WO2005/078843, WO2006/079800, WO2006/106334, WO2007/085863, WO2007/110587, WO2008/001119, WO2008/003976, WO2008/015461, WO2008/053213, WO2008/104760, WO2008/132493, WO2009/090419, WO2010/020797, WO2010/061190, WO2015/004419, WO2015/136295, WO2016/124929, WO2016/124928, WO2016/128721 and WO2016/083780. All publications referenced herein and their references are incorporated herein by reference in their entirety. Definition of terms used herein can be found as necessary in the above publications.

(3) Fuel cell stacks, fuel cell stack assemblies, fuel cell units (including fuel cell stack layers), and the arrangement of fuel cells within a fuel cell stack unit and fuel cell stack layer are well known.

(4) The present invention seeks to improve upon the prior art. The present invention seeks to improve fuel distribution and fuel flow within individual fuel cell units (fuel cell stack layers). In particular, it seeks to do one or more of: improve fuel velocity within fuel cell units, reduce fuel residence time, improve fuel distribution, and reduce the fuel pressure drop across fuel cell units.

SUMMARY OF THE INVENTION

(5) According to a first aspect of the present invention, there is provided a metal supported solid oxide fuel cell unit comprising: a) a metal substrate defining first and second opposed surfaces, wherein at least one solid oxide fuel cell is disposed on said second surface of said metal substrate; b) a metal spacer, which defines first and second opposed surfaces, said metal spacer comprising: (i) an external perimeter, (ii) at least one fuel inlet internal perimeter defining a fuel inlet port, (iii) at least one cut-out internal perimeter defining a cut-out, and (iv) at least one fuel outlet internal perimeter defining a fuel outlet port, wherein said first surface of said metal substrate is attached to said second surface of said metal spacer; and c) a metal interconnect plate which defines first and second opposed surfaces, said second surface of said metal interconnect plate sealingly attached to said first surface of said metal spacer,

(6) wherein: a fuel inlet port volume is defined between said first surface of said metal substrate, each at least one fuel inlet internal perimeter of said metal spacer, and said second surface of said metal interconnect plate, a cut-out volume is defined between said first surface of said metal substrate, said at least one cut-out internal perimeter of said metal spacer, and said second surface of said metal interconnect plate, and a fuel outlet port volume is defined between said first surface of said metal substrate, each at least one fuel outlet internal perimeter of said metal spacer, and said second surface of said metal interconnect plate, wherein said metal interconnect plate comprises a plurality of bridge portions defining a fluid flow path from said at least one fuel inlet port volume to said at least one cut-out volume to said at least one fuel outlet port volume.

(7) Thus, in embodiments comprising a plurality of fuel inlet internal perimeters defining fuel inlet ports, there are a plurality of fuel inlet port volumes.

(8) Thus, in embodiments comprising a plurality of cut-out internal perimeters defining cut-outs, there are a plurality of cut-out volumes.

(9) Thus, in embodiments comprising a plurality of outlet port internal perimeters defining fuel outlet ports, there are a plurality of fuel outlet port volumes.

(10) Thus, each fuel inlet port, each cut-out, and each fuel outlet port defined in the metal spacer are separate from one another, i.e. are divided from one another. Thus, the internal perimeters define dividers in the metal spacer between one another. They can also be described as being partitioned from one another, and discontinuous. Preferably, a fluid flow passage is not defined in the metal substrate between any of the fuel inlet ports, the cut-outs, and the fuel outlet ports.

(11) Preferably, the metal spacer is generally flat, i.e. is generally planar. Preferably, the metal substrate is generally flat, i.e. is generally planar.

(12) The bridge portions act to define a fluid flow bridge or channel (i.e. a fluid flow path) between adjacent volumes, e.g. (a) between fuel inlet port volumes and adjacent cut-out volumes, (b) between adjacent cut-out volumes, and (c) between cut-out volumes and fuel outlet port volumes. Thus, a fluid flow path is defined from the at least one fuel inlet port to the at least one cut-out to the at least one fuel outlet port via the bridge portions.

(13) Preferably there are a plurality of bridge portions between adjacent volumes. Preferably there are at least three bridge portions, more preferably at least four, between adjacent volumes, more particularly between adjacent volumes in the direction fuel inlet port to fuel outlet port. Preferably there are a plurality of bridge portions between a fuel inlet port volume and a cut-out volume, more preferably between each. Preferably, there are a plurality of bridge portions between a cut-out volume and a fuel outlet port volume, more preferably between each. Preferably there are a plurality of bridge portion between adjacent cut-out volumes, more preferably between each. More preferably, there are a plurality of bridge portions between adjacent cut-outs in a path from a fuel inlet port volume to a fuel outlet port volume, more particularly between adjacent cut-outs in the direction fuel inlet port to fuel outlet port, or in a direct or shortest path.

(14) The provision of bridge portions in the metal interconnect plate together with fuel inlet internal perimeters in the metal spacer allows the provision of complex fuel inlet and outlet ports which can provide for managed fluid flow distribution from the at least one fuel inlet port to the at least one cut-out, and from the at least one cut-out to the at least one fuel outlet port. This can e.g. allow for a more even distribution of fuel within the fuel cell unit, and thus help optimise fuel cell unit operation.

(15) Preferably, the bridge portions extend outwardly from the first surface of the metal interconnect plate, away from the second surface of the metal interconnect plate. More preferably, the bridge portions comprise a protrusion from the first surface of the metal interconnect plate. More preferably, the bridge portions comprise an indent in the second surface of the metal interconnect plate and a corresponding protrusion from the first surface of the metal interconnect plate. More preferably, the bridge portions comprise a dimple. More preferably, the bridge portions comprise an elongate dimple. Thus, the bridge portions may for example have a lozenge shape. Other shapes will be readily apparent to one of ordinary skill in the art.

(16) Preferably, the bridge portions define a volume between the first surface of the metal spacer and the second surface of the metal interconnect plate.

(17) Preferably, the metal interconnect plate comprises at least one fuel inlet bridge portion defining a fluid flow channel between a fuel inlet port volume and a cut-out volume, and at least one fuel outlet bridge portion defining a fluid flow channel between a cut-out volume and a fuel outlet port volume. In embodiments comprising a plurality of cut-outs, preferably the metal interconnect plate comprises at least one cut-out bridge portion defining a fluid flow channel between adjacent cut-outs.

(18) Preferably, the metal spacer comprises at least two fuel inlet internal perimeters defining at least two fuel inlet ports. Preferably, the metal spacer comprises at least two cut-out internal

perimeters defining at least two cut-outs. Preferably, the metal spacer comprises at least two fuel outlet internal perimeters defining at least two fuel outlet ports.

(19) Preferably, each metal spacer fuel port (each fuel inlet port and each fuel outlet port) comprises a fuel duct region, a plurality of fuel throat regions, and a corresponding plurality of fuel distributor channel regions (i.e. each fuel throat region leading to a fuel distributor channel region). Preferably, the fuel duct regions of the metal spacer fuel ports are aligned with the fuel ports of the metal substrate and the metal interconnect plate.

(20) The provision of the plurality of fuel throat regions and fuel distributor channel regions also reduces the risk of fuel starvation due to blockage.

(21) Preferably, there is at least one bridge portion between each fuel distributor channel region and the adjacent cut-out.

(22) CFD (computational flow dynamics) analysis shows that (compare e.g. FIGS. 16B (prior art) and 16C (present invention)), the fuel velocity within the fuel cell unit (particularly, across the at least one cut-out volume) remains more constant in the present design as compared to prior art products, with less fuel deficiency regions at the corner of the fuel cell, and the fuel promoted in a more uniform manner across said cell, which improve the chemical reaction which occurs within the fuel cell.

(23) CFD analysis also shows that the normalised fuel residence time is decreased as compared to prior art devices (compare e.g. FIGS. 17A (prior art) and 17B (present invention)). This means that a lower concentration of Hydrogen is required for the chemical reaction at the fuel cell. That is to say, less fuel is required for the chemical reaction to occur at the fuel cell, and therefore the present invention is more efficient than prior art devices.

(24) CFD analysis also shows improved flow distribution across the active region of the fuel cell as compared to prior art devices. CFD analysis also shows that the pressure drop between inlet and outlet ports is less (i.e. is improved) with the present invention as compared to prior art devices. Minimizing the pressure drop across the cell is beneficial to maintain the compression along the stack.

(25) The plurality of fuel throats are dimensioned to restrict (i.e. control) fluid flow from the fuel duct region. Preferably, each fuel distributor channel region is wider than (i.e. has a larger cross-section than) its corresponding fuel throat. Preferably, the fuel throat defines a fluid flow axis, and the width (cross-section) of the fuel throat perpendicular to the fluid flow axis is less than the width (cross-section) of the corresponding fuel distributor channel perpendicular to the fluid flow axis.

(26) Preferably, each fuel throat region has a constant width “W” between the fuel duct region and the corresponding fuel distributor channel region. In use, this allows fuel to be transferred at high speed (i.e. high velocity) from the fuel duct region to the corresponding fuel distributor channel region, and this reduces the risk of fuel starvation.

(27) In certain embodiments, all fuel throat regions have the same width.

(28) Preferably, each fuel distributor channel region has a width which increases from the fuel distributor channel proximal (closest to) the fuel throat region to the end of the fuel distributor channel proximal the adjacent cut-out internal perimeter. Preferably, each fuel distributor channel region has a curved shape being the width of said fuel distributor channel region. More preferably, this increases, starting from the width “W” of the fuel throat region and finishing at a width equal to a distance (d.sub.a, d.sub.b, d.sub.c), being (d.sub.a < d.sub.b < d.sub.c).

(29) Preferably, each cut-out internal perimeter (and each cut-out defined by it) has a plurality of corner regions. Fuel flow to corner regions of fuel cells is known to be difficult—it is highly desirable to optimise fuel flow to those corner regions. Preferably, with metal spacer fuel port comprising a plurality of fuel throats and corresponding fuel distributor channel regions, the fuel distributor channel region closest to (i.e. adjacent, or proximal) a corner region (or corner) has a final width (a width at the point closest to or proximal the adjacent cut-out internal perimeter) less than that of the other fuel distributor channel regions. More preferably, with a plurality of fuel

distributor channel regions arranged in the order: fuel distributor channel region closest to a corner region (i.e. corner) to fuel distributor channel region furthest from the corner region (i.e. corner), each fuel distributor channel region has a final width greater than that of the preceding fuel distributor channel region.

(30) Preferably, the width (or distance) d at the edge of the fuel distributor channel region adjacent the adjacent cut-out internal perimeter is greater than (i.e. longer) in the areas close to the centre of the cell to promote the fuel uniformly along the middle region of the cell, improving upon the fuel distribution of previous designs wherein there were problems of fuel deficiency in said central region.

(31) Preferably, each fuel throat region has a length (“L”). More preferably, in fuel cell stacks and fuel cell stack assemblies of the present invention (below), the fuel throat region **44b** is related to the size of the compression gaskets located in between the interconnect of one solid oxide fuel cell unit and the substrate layer of the following solid oxide fuel cell unit. More preferably, the gaskets comprise a toroid shape surrounding a port of interconnect. The length “L” of the fuel throat region is preferably coincident with (corresponds with, is related to) the external radius minus the internal radius of the compression gasket. Such arrangements may assist in reducing or minimising pressure drop.

(32) Preferably, the fuel used in the products of the present invention is a hydrocarbon fuel. Suitable hydrocarbon fuels include desulfurized hydrocarbon fuel, reformat or reformat mixed with an anode off-gas (i.e. an anode-side exhaust gas from a fuel outlet side of a fuel cell unit). Similarly, the fuel used can be start up or shut down gases from a fuel cell unit, more particularly start up or shut down anode off-gas (i.e. an anode-side exhaust gas from a fuel outlet side of a fuel cell unit).

(33) The terms “disposed on” and “attached to” are used interchangeably herein.

(34) Preferably, the metal supported solid oxide fuel cell unit is a fuel cell stack layer, more preferably a metal supported solid oxide fuel cell stack layer. Thus, a plurality of fuel cell units may be assembled to form a solid oxide fuel cell stack.

(35) Preferably, the metal substrate (or in embodiments where the metal spacer comprises at least one metal spacer plate), each metal substrate plate (also referred to as a “fuel cell plate”) comprises at least one porous region. Preferably, the at least one porous region is surrounded by a non-porous region. More preferably, each metal substrate or each metal substrate plate comprises one porous region. More preferably, each porous region is surrounded by a non-porous region. More preferably, the or each porous region is a perforated region. Preferably, it comprises (i.e. is defined by) a plurality of perforations extending from the first surface to the second surface (i.e. between the first surface and the second surface). More preferably, the perforations are laser drilled perforations. Preferably, the at least one non-porous region of the metal spacer or each metal substrate plate is attached to the metal spacer. Preferably, each at least one cut-out internal perimeter is wholly overlapped by said metal substrate.

(36) Preferably, the at least one porous region is coincident with (i.e. extends to, or overlaps) the corresponding cut-out internal perimeter of the metal substrate, i.e. extends to the bound of the cut-out internal perimeter. This is particularly preferable in embodiments where the metal substrate plate comprises a single porous region.

(37) Preferably, each solid oxide fuel cell disposed on a metal substrate plate comprises an anode layer deposited over (bonded to) a porous region of the metal substrate plate, an electrolyte layer deposited over (bonded to) the anode layer, and a cathode layer deposited over the electrolyte layer. Preferably, the electrolyte layer extends over the anode to sealingly attach to the non-perforated region of the metal substrate plate surrounding the anode.

(38) In certain embodiments, the metal substrate is provided as a single component. In other embodiments, the metal substrate is provided as a plurality of discrete components.

(39) In certain embodiments, the metal substrate comprises at least one metal substrate plate (more

preferably, at least two metal substrate plates) and at least two blanking plates, each metal substrate plate defining first and second opposed surfaces and each blanking plate defining first and second opposed surfaces, wherein at least one solid oxide fuel cell is disposed on said second surface of each metal substrate plate, and wherein said first surface of each metal substrate plate and said first surface of each blanking plate is attached to said second surface of said metal spacer, each at least one cut-out internal perimeter of said metal spacer being wholly overlapped by a metal substrate plate.

(40) Thus, the plurality of metal substrate plates attached to the metal spacer together define a metal substrate (also referred to as a “substrate layer” or “metal substrate layer”) attached to the metal spacer. Thus, each fuel cell unit comprises a metal substrate, a metal spacer, and a metal interconnect plate.

(41) The metal substrate plates are preferably disposed between and on the same plane as the blanking plates. The blanking plates are preferably of a generally rectangular shape. Preferably, each blanking plate comprises at least one internal perimeter defining a fuel port, i.e. defining at least one fuel port. More preferably, each blanking plate defines two fuel ports.

(42) Preferably, the metal substrate plates and the blanking plates do not contact one another, i.e. do not abut one another. Thus, preferably a blanking plate (such as a first blanking plate) does not abut or contact an adjacent metal substrate plate (such as a first metal substrate plate). Thus, preferably adjacent metal substrate plates do not abut or contact one another. Thus, preferably a second blanking plate does not abut or contact an adjacent second metal substrate plate.

(43) By having the blanking plates and metal substrate plates attached to the metal spacer, and not having them abutting or contacting one another, this allows for a “tolerance gap” to be defined between them, and allows for variability in the positioning of the components during the manufacturing process. This provides a significant technical advantage in terms of the manufacturing process and can e.g. assist in an increased manufacturing process speed, reduced cost, increased reliability of the fuel cell unit, and/or increased lifespan of the fuel cell unit.

(44) Preferably, the blanking plates are metal blanking plates. More preferably, the blanking plates are made of the same metal as the metal substrate plates. Preferably, the blanking plates are of equal thickness to the metal substrate plates.

(45) Preferably, the metal spacer comprises at least two internal perimeters each defining a fuel port. More preferably, each metal spacer defines two fuel ports at a first end (preferably, a fuel inlet end, i.e. defines at least two fuel inlet ports), and two fuel ports at a second end (preferably, an exhaust fuel outlet end, i.e. defines at least two fuel outlet ports). The cut-out internal perimeters defining the cut-outs may be considered a first set of internal perimeters, and the internal perimeters defining the fuel ports may be considered a second set of internal perimeters.

(46) Each metal substrate plate is attached wholly overlapping a cut-out internal perimeter i.e. on and over a cut-out. Thus, each metal substrate plate covers a cut-out. Each metal substrate plate is attached to the metal spacer between an at least one internal perimeter defining a cut-out and the external perimeter of the metal spacer.

(47) Preferably, the metal substrate is attached (or the at least two blanking plates and the at least one metal substrate plate) are attached to the metal spacer by welding, more preferably by line welding.

(48) Preferably, each metal substrate plate is attached to the metal spacer between an at least one cut-out internal perimeter defining a cut-out and the external perimeter of the metal spacer. More preferably, each metal substrate plate is attached to the metal spacer between a cut-out internal perimeter defining a cut-out and the external perimeter of the metal spacer.

(49) Preferably, the metal interconnect plate is sealingly attached to the metal spacer by welding.

(50) Preferably, the metal interconnect plate is sealingly attached to the metal spacer along a line disposed between (a) the external perimeter of the metal spacer, and (b) the at least one cut-out internal perimeter of the metal spacer. Preferably, the line is proximal the external perimeter of the

metal spacer. More preferably, the line is adjacent the external perimeter of the metal spacer. More preferably, the line is within 10 mm, more preferably within 5 mm, more preferably within 4 mm, more preferably within 3 mm, more preferably within 2 mm of the external perimeter of the metal spacer.

(51) More preferably, the weld(s) attaching the metal substrate (or the at least two blanking plates and the at least one metal substrate plate) to the metal spacer, and the weld sealingly attaching the metal interconnect plate to the metal spacer do not overlap.

(52) Preferably, the first surface of each metal substrate plate and the first surface of each blanking plate is disposed on and attached to the second surface of the metal spacer.

(53) Preferably, the second surface of the metal interconnect plate is disposed on and sealingly attached to the first surface of the metal spacer.

(54) Preferably, there are a total of two blanking plates.

(55) Preferably, the metal interconnect plate comprises a plurality of dimples extending outwardly from the first surface, away from the second surface. Preferably, the metal interconnect plate comprises a plurality of dimples extending outwardly from the first surface, away from the second surface, and a plurality of dimples extending away from the first surface, outwardly from the second surface. Preferably, the dimples alternate. Thus, preferably, the dimples alternate between extending away from the first surface and from the second surface of the metal interconnect plate.

(56) Preferably, the fuel cell unit comprises at least one combined metal substrate plate, each combined metal substrate plate comprising a (combined) blanking plate and at least one metal substrate plate. Thus, a base plate and at least one metal substrate plate are attached to one another to form a single component (a combined metal substrate plate), or are integrally formed as a single component from the outset, prior to the combined metal substrate plate being attached to the metal spacer. More preferably, the fuel cell unit comprises two combined metal substrate plates. More preferably, the fuel cell unit comprises a combined metal substrate plate, at least one metal substrate plate, and a blanking plate. Alternatively, the fuel cell unit comprises two combined metal substrate plates, and at least one metal substrate plate.

(57) In embodiments comprising at least one combined metal substrate plate, a combined metal substrate plate preferably does not abut or contact an adjacent metal substrate plate or combined metal substrate plate.

(58) As well as providing metal substrate plates in a 2×1 series (linear) arrangement between the blanking plates, other arrangements and numbers of metal substrate plates can be provided. For example, metal substrate plates may be provided in a 1×2 (parallel) arrangement between the blanking plates. Alternatively, metal substrate plates may be provided in a 2×2, 3×2 or 4×2 arrangement and the same blanking plates may be used with the various arrangements. Similarly, fuel cell units may be provided with metal substrate plates in 2×3, 3×3 or 4×3 arrangements using the same blanking plates. Other arrangements will be readily apparent.

(59) The use of the plurality of metal substrate plates allows the metal substrate plates to be combined in a modular manner to obtain a range of fuel cell unit sizes and a range of power outputs as desired, including e.g. a larger fuel cell unit size and therefore larger power output. The attachment of the metal substrate plates to the metal spacer can also reduce the likelihood of bending of the fuel cells within the fuel cell unit, therefore decreasing the risk of reduced electrical conductivity and reduced gas sealing that may occur should the fuel cell bend. The use of metal substrate plates also means that a given metal substrate plate can be manufactured and used in a number of different fuel cell unit products. Thus, for example, it could be used in a fuel cell unit which contains just two metal substrate plates. Alternatively, it could be used in a larger fuel cell unit, such as one having 4, 6, 8, 9, 10 or 12 metal substrate plates. This can act to reduce the cost and increase the speed, quality and reliability of manufacture of the metal substrate plate (and its at least one fuel cell).

(60) In various embodiments, two blanking plates are provided, arranged (when attached to the

metal spacer) with one at either end of the metal substrate, i.e. one at a first end of the metal substrate, and the other at a second end of the metal substrate. In other embodiments, blanking plates may be split into first and second portions i.e. a first blanking plate portion and a second blanking plate portion. Thus a first blanking plate can comprise a first blanking plate portion and a second blanking plate portion. Similarly, a second blanking plate can comprise a first blanking plate portion and a second blanking plate portion, each blanking plate portion attached to the metal spacer.

(61) The fuel cell unit has internal fuel manifolding. When a plurality of the fuel cell units are assembled into a stack an open manifolded oxidant (air) flow passage is defined between (a) the first surface of the metal interconnect plate of a first fuel cell unit, and (b) the second surfaces of the metal substrate (or the at least two blanking plates and the at least one metal substrate plate) of an adjacent second fuel cell unit. Preferably, the metal interconnect plate comprises a plurality of dimples extending outwardly from the first surface, away from the second surface. Preferably, the dimples of a first fuel cell unit abut the cathode layer of the at least one solid oxide fuel cell of an adjacent second fuel cell unit and act as a current collector.

(62) The above optional and preferable features are equally applicable to other aspects of the present invention detailed below.

(63) Also provided according to the present invention is a solid oxide fuel cell stack comprising a plurality of metal supported solid oxide fuel cell units according to the present invention.

(64) Preferably, the solid oxide fuel cell units are separated from one another by compression gaskets.

(65) Also provided according to the present invention is a solid oxide fuel cell stack assembly comprising: a base plate, an end plate, a solid oxide fuel cell stack according to the present invention positioned between the base plate and the end plate, and a skirt attached to the base plate and the end plate and defining a volume between the skirt, the base plate and the end plate within which is contained the fuel cell stack.

(66) Also provided according to the present invention is a method of assembly of a metal supported solid oxide fuel cell unit, the metal supported solid oxide fuel cell unit comprising: a) a metal substrate defining first and second opposed surfaces, wherein at least one solid oxide fuel cell is disposed on said second surface of said metal substrate; b) a metal spacer, which defines first and second opposed surfaces, said metal spacer comprising: (i) an external perimeter, (ii) an at least one fuel inlet internal perimeter defining a fuel inlet port, (iii) at least one cut-out internal perimeter defining a cut-out, and (iv) at least one fuel outlet internal perimeter defining a fuel outlet port; and c) a metal interconnect plate which defines first and second opposed surfaces, said method of assembly comprising the steps of: (i) attaching said first surface of said metal substrate to said second surface of said metal spacer; and (ii) sealingly attaching said second surface of said metal interconnect to said first surface of said metal spacer

(67) wherein: a fuel inlet port volume is defined between said first surface of said metal substrate, each at least one fuel inlet internal perimeter of said metal spacer, and said second surface of said metal interconnect plate, a cut-out volume is defined between said first surface of said metal substrate, each at least one cut-out internal perimeter of said metal spacer, and said second surface of said metal interconnect plate, and a fuel outlet port volume is defined between said first surface of said metal substrate, each at least one fuel outlet internal perimeter of said metal spacer, and said second surface of said metal interconnect plate, wherein said metal interconnect plate comprises a plurality of bridge portions defining a fluid flow path from said at least one fuel inlet port volume to said at least one cut-out volume to said at least one fuel outlet port volume.

(68) In embodiments where the metal substrate comprises at least two blanking plates and at least one metal substrate plate (preferably, a plurality of metal substrate plates), preferably, step (i) comprises clamping the metal spacer to the at least two blanking plates and the at least one metal substrate plate and attaching the metal spacer to the at least two blanking plates and the at least one

metal substrate plate.

(69) Preferably, step (ii) comprises clamping the metal interconnect plate to the metal spacer and attaching the metal interconnect plate to the metal spacer.

(70) Preferably, at least one of step (i) and step (ii) comprises attaching by welding. More preferably, both steps (i) and (ii) comprise attaching by welding.

(71) Preferably, the at least one metal substrate plate and the plurality of blanking plates is aligned with the metal spacer and aligned with the metal interconnect plate.

(72) Preferably, locating means (also referred to as positioning means) are used to locate the various components during the assembly process. Suitable location means include datum edges, fixed dowels, and spring dowels. Other location means will be readily apparent to one of ordinary skill in the art.

(73) Preferably, the solid oxide fuel cell unit is assembled by attaching the metal substrate (or the at least one metal substrate plate and blanking plates) to the metal spacer so that the metal substrate (or the at least one metal substrate plate) is attached over the at least one cut out in the metal spacer. The metal substrate (or the at least two blanking plates and at least one metal substrate plate) and the metal spacer are preferably clamped together using a first clamping plate. Preferably, the metal substrate (or the at least two blanking plates and the at least one metal substrate plate) are positioned on a base plate, and the metal spacer plate is positioned on top. Preferably, a first clamping plate is positioned over the metal spacer. More preferably, clamping means clamps the metal substrate (or the at least two blanking plates and the at least one metal substrate plate) and the metal spacer between the base plate and the first clamping plate. More preferably, the first clamping plate defines welding slots through which the metal substrate (or the at least two blanking plates and the at least one metal substrate plate) is/are welded to the metal spacer.

(74) Preferably, the metal interconnect plate is attached to the metal spacer by welding. Preferably, the metal interconnect plate is placed over the metal spacer to which the metal substrate has already been attached (or the at least one metal substrate plate and blanking plates have already been attached). Preferably, a second clamping plate is positioned over the metal interconnect plate. More preferably, clamping means clamps the metal substrate (or the at least two blanking plates and at least one metal substrate plate), the metal spacer, and the metal interconnect plate between the base plate and the second clamping plate. Preferably, the second clamping means defines an opening. More preferably, the metal interconnect plate is welded to the metal substrate through the opening. Preferably, the welding is between the external perimeters of the metal substrate and the metal interconnect plate, and the internal perimeters of the metal spacer. More preferably, the welding extends through the metal interconnect plate, metal substrate, and through to the metal substrate (or the at least two blanking plates and the at least one metal substrate plate).

(75) The at least two blanking plates and the at least one metal substrate plate attached to the metal spacer together define a metal substrate.

(76) Preferably, the metal substrate (or the at least two blanking plates, the at least one metal substrate plate), the metal spacer, and the metal interconnect plate are aligned during assembly by means of a datum edge.

(77) An enabling disclosure of the present invention, to one of ordinary skill in the art, is provided herein. Reference now will be made in detail to embodiments of the invention, one or more examples of which are set forth below. Each example is provided by way of explanation of the invention, not limitation of the invention.

Description

(1) Of the Figures:

(2) FIG. 1 shows an exploded perspective view of the fuel cell unit components of Embodiment 1

- (3) FIG. 2 shows a top view of metal substrate components located on an assembly baseplate
- (4) FIG. 3 shows a top view of a metal spacer positioned on top of metal substrate components of FIG. 2
- (5) FIG. 4 shows a top view of a first clamping plate positioned on top of the metal spacer of FIG. 3 for welding purposes
- (6) FIG. 5 shows a top view of the metal spacer of FIG. 3 after welding and removal of the first clamping means
- (7) FIG. 6 shows a top view of a metal interconnect plate positioned on top of the metal spacer of FIG. 5
- (8) FIG. 7 shows a top view of a second clamping plate positioned on top of the metal interconnect plate of FIG. 6 for welding purposes
- (9) FIG. 8 shows a top view of the metal interconnect plate of FIG. 6 after welding and removal of the second clamping means and removal from the assembly base plate
- (10) FIG. 9 shows a cross-section through a metal substrate plate
- (11) FIG. 10 shows an exploded perspective view of the fuel cell unit of Embodiment 2
- (12) FIG. 11 shows an exploded perspective view of the fuel cell unit of Embodiment 4
- (13) FIG. 12 shows a top view of the component parts of the fuel cell unit of the fuel cell unit of Embodiment 5
- (14) FIG. 13 shows a top view of a part of a metal spacer
- (15) FIG. 14 shows a top view of a part of a metal interconnect plate
- (16) FIG. 15 shows an exploded perspective view of a fuel cell unit of Embodiment 1 with an illustration of fuel flow
- (17) FIG. 16A is a CFD image showing fuel velocity in a prior art device
- (18) FIG. 16B is a CFD image showing fuel velocity in a prior art device (shading indicates fuel mid plane velocities in $m \cdot s^{-1}$)
- (19) FIG. 16C is a CFD image showing fuel velocity in a device according to the present invention (shading indicates fuel mid plane velocities in $m \cdot s^{-1}$)
- (20) FIG. 17A is a CFD image showing normalised fuel residence time (age of fuel) in a prior art device
- (21) FIG. 17B is a CFD image showing normalised fuel residence time (age of fuel) in a device according to the present invention
- (22) A list of the reference signs used herein is given at the end of the specific embodiments. Repeat use of reference symbols in the present specification and drawings is intended to represent the same or analogous features or elements.
- (23) It will be apparent to those of ordinary skill in the art that various modifications and variations can be made in the present invention without departing from the scope of the appended claims. For instance, features described as part of one embodiment can be used on another embodiment to yield a still further embodiment. Thus, it is intended that the present invention cover such modifications and variations as come within the scope of the appended claims and their equivalents.
- (24) Other objects, features, and aspects of the present invention are disclosed in the remainder of the specification. It is to be understood by one of ordinary skill in the art that the present discussion is a description of exemplary embodiments only and is not intended as limiting the broader aspects of the present invention, which broader aspects are embodied in the exemplary constructions.

EMBODIMENT 1

- (25) Fabrication of a metal supported solid oxide fuel cell unit **1** is illustrated in the Figures. Metal supported solid oxide fuel cell unit **1** is for use as a solid oxide fuel cell stack layer.
- (26) In this embodiment, metal supported solid oxide fuel cell unit **1** is fabricated comprising a metal substrate **65** (also referred to as a “substrate layer” or a “metal substrate layer”), a metal spacer **30**, and a metal interconnect plate **20**.
- (27) Metal substrate plates **70a** and **70b** each comprise a porous region **78** defined by laser-drilled

perforations **78a** extending between first surface **71** and second surface **72**. Fuel cell **79** is deposited over porous region **78** on second surface **72** of metal substrate plates **70a** and **70b**, and comprises an anode layer deposited over (bonded to) porous region **78** of metal substrate plate **70a**, **70b**, an electrolyte layer deposited over (bonded to) the anode layer, and a cathode layer deposited over the electrolyte layer. Porous region **78** is surrounded by non-porous region **78b**.

(28) As shown in FIG. 2, assembly baseplate **80** comprises fixed dowels **83a**, **83b**, **83c**, **83d**, **83e**, **83f**, **83g**, and spring loaded dowels **84a**, **84b**, **84c**, **84d**, **84e**, **84f**, **84g**. Assembly baseplate **80** also defines (comprises) a datum edge **81**.

(29) Metal substrate plates **70a** and **70b**, and blanking plates **50a** and **50b** are aligned on assembly baseplate **80**, and alignment is achieved by fixed dowels **83a**, **83b**, **83c**, **83d**, **83e**, **83f**, **83g**, spring loaded dowels **84a**, **84b**, **84c**, **84d**, **84e**, **84f**, **84g**, and datum edge **81**.

(30) Second surface **52** of blanking plate **50a** is disposed on (i.e. contacts/abuts) assembly baseplate **80**. Second edge **58** of blanking plate **50a** is aligned on datum edge **81** by fixed dowel **83g**, and first edge **57** of blanking plate **50a** is aligned to fixed dowel **83a** and spring loaded dowel **84a**. Curved edge **55** of blanking plate **50a** is aligned by spring loaded dowel **84g**.

(31) Second surface **52** of blanking plate **50b** is disposed on (i.e. contacts/abuts) assembly baseplate **80**. Second edge **58** of blanking plate **50b** is aligned on the datum edge **81** by fixed dowel **83c**, and first edge **57** of blanking plate **50b** is aligned to fixed dowel **83b** and spring loaded dowel **84d**. Curved edge **55** of blanking plate **50b** is aligned with spring loaded dowel **84e**.

(32) Second surface **72** of metal substrate plate **70a** is disposed on (i.e. contacts/abuts) assembly baseplate **80**.

(33) Metal substrate plates **70a** and **70b** are positioned on assembly baseplate **80** between blanking plates **50a** and **50b**. Second short side **75** of metal substrate plate **70a** is aligned on datum edge **81** by fixed dowels **83f** and **83e**. First short side **74** of metal substrate plate **70a** is aligned by spring loaded dowel **84b**.

(34) Second short side **75** of metal substrate plate **70b** is aligned on datum edge **81** by fixed dowel **83d** and spring loaded dowel **84f**. First short side **74** of metal substrate plate **70b** is aligned by spring loaded dowel **84c**.

(35) Outer long side **76** of metal substrate plate **70a** is aligned parallel to inner edge **59** of the blanking plate **50a**, defining a tolerance gap **82a** between metal substrate plate **70a** and blanking plate **50a**.

(36) Outer long side **76** of metal substrate plate **70b** is aligned parallel to inner edge **59** of blanking plate **50b**, defining a tolerance gap **82b** between metal substrate plate **70b** and blanking plate **50b**.

(37) Tolerance gap **82c** is defined between inner long side **77** of metal substrate plate **70a** and inner long side **77** of metal substrate plate **70b**.

(38) As shown in FIG. 3, metal spacer **30** is then placed on top of blanking plate **50a**, metal substrate plate **70a**, metal substrate plate **70b** and blanking plate **50b**.

(39) Second surface **32** of metal spacer **30** is disposed on (i.e. contacts/abuts) first surface **51** of blanking plate **50a**, first surface **71** of metal substrate plate **70a**, first surface **71** of metal substrate plate **70b**, and first surface **51** of blanking plate **50b**.

(40) Metal spacer **30** is aligned with blanking plate **50a**, metal substrate plate **70a**, metal substrate plate **70b** and blanking plate **50b** by fixed dowel **83e**, spring loaded dowels **84a**, **84d**, **84e**, **84f** and **84g**, and datum edge **81**.

(41) Second elongate edge **38** of metal spacer **30** is aligned with datum edge **81** and second edge **58** of blanking plates **50a** and **50b** and second short sides **75** of metal substrate plates **70a** and **70b** using fixed dowel **83e** and spring loaded dowel **84f**. First elongate edge **37** of metal spacer **30** is aligned with first edges **57** of blanking plates **50a** and **50b** and first short sides **74** of metal substrate plates **70a** and **70b** using spring loaded dowels **84a** and **84d**.

(42) The external perimeters of blanking plate **50a**, metal substrate plate **70a**, metal substrate plate **70b** and blanking plate **50b** do not extend beyond external perimeter **33** of metal spacer **30**.

(43) Metal spacer **30** comprises cut-out internal perimeters **39a** and **39b**, with each internal perimeter defining a respective cut-out **40a** and **40b**, and a cross member **41** between them. Metal substrate plates **70a** and **70b** wholly overlap cut-out internal perimeters **39a** and **39b** of metal spacer **30**, i.e. metal substrate plates **70a** and **70b** wholly cover cut-outs **40a** and **40b**.

(44) Metal spacer **30** also comprises a plurality of fuel inlet internal perimeters **33a**, **33b**, and fuel outlet internal perimeters **33c**, **33d** defining fuel ports **34a**, **34b**, **34c** and **34d**. Each fuel port comprises a number of regions—fuel duct region **44a**, fuel throat region **44b**, and fuel distributor channel region **44c**.

(45) As shown in FIG. **4**, first clamping plate **90** is then placed on top of metal spacer **30**, i.e. contacts/abuts first surface **31** of metal spacer **30**.

(46) First clamping plate **90** defines orifices **92a** and **92b**. Spring loaded dowel **84h** projects through orifice **92a**, and fixed dowel **83e** projects through orifice **92b**, allowing first clamping plate **90** to be aligned with metal spacer **30** (and therefore also with blanking plate **50a**, metal substrate plate **70a**, metal substrate plate **70b** and blanking plate **50b**).

(47) Clamping means (not shown) clamps first clamping plate **90** and assembly baseplate **80**, i.e. clamps metal spacer **30**, blanking plate **50a**, metal substrate plate **70a**, metal substrate plate **70b** and blanking plate **50b**.

(48) First clamping plate also defines welding slots **91a**, **91b** and **91c**.

(49) Welding means (not shown) is used to create line weld seam **100a** between metal spacer **30** and blanking plate **50a**, line weld seams **100b** and **100c** between metal spacer **30** and metal substrate plate **70a**, line weld seams **100d** and **100e** between metal spacer **30** and metal substrate plate **70b**, and line weld seam **100f** between metal spacer **30** and blanking plate **50b**.

(50) Non-porous region **78b** of metal substrate plates **70a**, **70b** is attached to metal spacer **30**.

(51) Blanking plate **50a**, metal substrate plate **70a**, metal substrate plate **70b**, and blanking plate **50b** attached to metal spacer **30** form/define a metal substrate **65**, i.e. a metal substrate **65** attached to metal spacer **30**.

(52) First clamping plate **90** is then removed, as shown in FIG. **5** (fixed dowel **83e** and spring loaded dowels **84a**, **84d** and **84f** are not shown).

(53) As shown in FIG. **6**, metal interconnect plate **20** is then placed on top of metal spacer **30**.

(54) Second surface **22** of metal interconnect plate **20** is disposed on (i.e. contacts/abuts) first surface **31** of metal spacer **30**.

(55) Metal interconnect plate **20** is aligned with metal spacer **30** (and therefore also blanking plate **50a**, metal substrate plate **70a**, metal substrate plate **70b** and blanking plate **50b**) by fixed dowel **83e**, spring loaded dowels **84a**, **84d**, and **84f**, and datum edge **81**. Spring loaded dowels **84a** and **84d** abut first edge **27** of metal interconnect plate **20**. Second edge **28** of metal interconnect plate **20** abuts datum edge **81**, fixed dowel **83e**, and spring loaded dowel **84f**.

(56) Metal interconnect plate **20** comprises a plurality of dimples **110** and elongate bridge dimples **120**, **121** which extend outwardly from first surface **21**, i.e. away from second surface **22** and away from metal spacer **30** and the metal substrate **65** attached to metal spacer **30**.

(57) Dimples **110** are formed in a number of regions including regions corresponding to the location of fuel cells **79** of metal substrate plates **70a**, **70b**, such that in a fuel cell stack arrangement comprising a plurality of fuel cell units **1** in a stack, the dimples **110** of a first fuel cell unit **1** contact the fuel cells **79** of an adjacent fuel cell unit **1** with which it is stacked. Thus, the dimples **110** form an electrical connection with the outer (cathode) surface of the fuel cells **79**, with electrical current flowing from the first surface **21** of metal interconnect plate **20** to the cathode layer of the adjacent fuel cell/fuel cells **79** of the adjacent fuel cell unit **1**.

(58) As described in more detail later, elongate bridge dimples **120**, **121** act as fluid flow bridges between separate zones/areas/volumes of the final fuel cell unit **1**.

(59) As shown in FIG. **7**, second clamping plate **95** is then placed on top of metal interconnect plate **20**, i.e. contacts/abuts first surface **21** of metal interconnect plate **20**.

(60) Second clamping plate **95** defines orifices **98a** and **98b**. Spring loaded dowel **84h** projects through orifice **98a**, and fixed dowel **83e** projects through orifice **98b**, allowing second clamping plate **95** to be aligned with metal interconnect plate **20** (and therefore also with metal spacer **30**, blanking plate **50a**, metal substrate plate **70a**, metal substrate plate **70b** and blanking plate **50b**).

(61) Second clamping plate **95** comprises inner perimeter **96** which defines opening **96a**.

(62) Clamping means (not shown) clamps second clamping plate **95** and assembly baseplate **80**, i.e. clamps metal interconnect plate **20**, metal spacer **30**, blanking plate **50a**, metal substrate plate **70a**, metal substrate plate **70b** and blanking plate **50b**.

(63) Welding means (not shown) is used to create a continuous perimeter weld seam **101** between metal interconnect plate **20**, metal spacer **30**, and blanking plate **50a**, metal substrate plate **70a**, metal substrate plate **70b**, and blanking plate **50b**.

(64) Second clamping plate **95** is then removed, and the completed metal supported solid oxide fuel cell unit **1** is removed from assembly base plate **80**.

(65) In the completed metal supported solid oxide fuel cell unit **1**, fuel ducts **130** are defined by fuel ports **24**, fuel duct regions **44a** of fuel ports **34a-d**, all of which are aligned with one another. Fuel ducts **130** extend between first surface **21** of metal interconnect plate **20** and second surface **52** of blanking plates **50a**, **50b**.

(66) At first end **2** of fuel cell unit **1** (see e.g. FIG. **3**), first volumes (fuel inlet port volumes **35a**) are defined between first surface **51** of blanking plate **50a**, fuel inlet internal perimeters **33a**, **33b** of metal spacer **30**, and second surface **22** of metal interconnect plate **20**.

(67) A second volume (a cut-out volume **35b**) is defined between first surface **71** of metal substrate plate **70a**, cut-out internal perimeter **39a** of metal spacer **30**, and second surface **22** of metal interconnect plate **20**.

(68) A third volume (a cut-out volume **35b**) is defined between first surface **71** of metal substrate plate **70b**, cut-out internal perimeter **39b** of metal spacer **30**, and second surface **22** of metal interconnect plate **20**.

(69) At second end **3** of fuel cell unit **1**, fourth volumes (fuel outlet port volumes **35c**) are defined between first surface **51** of blanking plate **50b**, fuel outlet internal perimeters **33c** and **33d** of metal spacer **30**, and second surface **22** of metal interconnect plate **20**.

(70) At first end **2** of fuel cell unit **1**, elongate dimples **120** act to define a fluid flow passage between the first and second volumes, i.e. act as fluid flow bridges between the first and second volumes. The fluid flow bridges are the volumes between the elongate dimples **120** and the metal spacer **30**.

(71) Elongate dimples **121** act to define a fluid flow passage between the second and third volumes (i.e. between the adjacent cut-out volumes **35b**). The fluid flow bridges are the volumes between the elongate dimples **121** and the metal spacer **30**.

(72) At second end **3** of fuel cell unit **1**, elongate dimples **120** act to define a fluid flow passage between the third and fourth volumes, i.e. act as fluid flow bridges between the first and second volumes. The fluid flow bridges are the volumes between the elongate dimples **120** and the metal spacer **30**.

(73) Thus, a fluid flow path is defined (using the fuel inlet port volumes **35a**, cut-out volumes **35b**, fuel outlet port volumes **35c**, and fluid flow bridges) from: (1) fuel duct regions **44a** of fuel ports **34a**, **34b**, to (2) fuel throat regions **44b** of fuel ports **34a**, **34b**, to (3) fuel distributor channel regions **44c** of fuel ports **34a**, **34b**, to (4) elongate dimples **120** at first end **2** of fuel cell unit **1**, to (5) the second volume defined between first surface **71** of metal substrate plate **70a**, cut-out internal perimeter **39a** of metal spacer **30**, and second surface **22** of metal interconnect plate **20**, to (6) elongate dimples **121**, to (7) the third volume defined between first surface **71** of metal substrate plate **70b**, cut-out internal perimeter **39b** of metal spacer **30**, and second surface **22** of metal interconnect plate **20**, to (8) fuel distributor channel regions **44c** of fuel ports **34c**, **34d**, to (9) fuel throat regions **44b** of fuel ports **34c**, **34d**, to (10) fuel duct regions **44a** of fuel ports **34c**, **34d**.

(74) Thus, a fluid flow path **140** (i.e. a fuel flow path) is defined within fuel cell unit **1** from fuel ducts **130** at first end **2** to fuel ducts **130** at second end **3**.

(75) The fluid flow path **140** is illustrated in FIG. **15**.

(76) As can be seen from FIGS. **13** and **14**, metal interconnect plate **20** comprises one elongate bridge dimple **120a** which transfer the fuel from one fuel distributor channel **44c1** to the cut out **40a**. Likewise, metal interconnect plate **20** comprises two elongate bridge dimples **120b** which transfer the fuel from one fuel distributor channel **44c2** to the cut out **40a**. Furthermore, metal interconnect plate **20** comprises three elongate bridge dimples **120c** which transfer the fuel from one fuel distributor channel **44c3** to the cut out **40a**.

(77) The width of the elongate bridge dimples r_a , r_b and r_{b1} remain constant. However, the width r_c , r_{c1} and r_{c2} of the elongate bridge dimple (the length of the shortest side of the rectangular cross section shape of elongate bridge dimples **120c**) increases as the elongate bridge dimples **120c** get close to the centre of the fuel cell unit. That is to say, $r_{c2} > r_{c1} > r_c$, so that the flow area enclosed inside the elongate bridge dimples **120c** closer to the middle of the fuel cell is gradually greater, being the fuel promoted uniformly to the centre of the fuel cell avoiding fuel starvation in said centre of the fuel cell.

(78) FIG. **14** also illustrates the alternating dimples **110** and **122**, with dimples alternating between extending away from the first surface and from the second surface of the metal interconnect plate. For the avoidance of doubt, elongate dimples **120**, **120a**, **120b**, **120c** and **121** do not alternate and instead all extend from first surface **21** of metal interconnect plate **20**, away from second surface **22**.

(79) FIG. **13** illustrates the arrangement of the fuel ports **34a**, **34b**, **34c** and **34d** of the metal spacer **30**. The fuel throat region **44b** comprises a constant width “W” between the fuel port **34a** to the corresponding fuel distributor channel region **44c** (fuel distributor channel regions **44c1**, **44c2**, **44c3**) wherein the fuel is transferred at high speed from such fuel ports **34a** to the corresponding fuel distributor channel region, reducing the risk of fuel starvation.

(80) In addition, fuel distributor channel region **44c** (**44c1**, **44c2**, **44c3**) comprises a curved shape being the width of said fuel distributor channel region **44c** gradually increasing, starting from the width “W” of fuel throat region **44b** and finishing at a width equal to a distance (d_a , d_b , d_c), wherein ($d_a < d_b < d_c$).

(81) The distance d (d_a , d_b , d_c) at the edge of the fuel distributor channel region **44c** is longer in the areas close to the centre of the fuel cell unit **1** to promote the fuel uniformly along the middle region of the fuel cell unit **1**, improving the fuel distribution.

(82) In a fuel cell stack assembly, the length “L” of the fuel throat region **44b** is related to the size of the compression gaskets located in between the metal interconnect plate **20** of one solid oxide fuel cell unit **1** and the substrate layer **65** of the following solid oxide fuel cell unit **1**, the gasket comprising a toroid shape surrounding one port of interconnect. The length “L” of the fuel throat region **44b** is coincident with the external radius minus the internal radius of the compression gasket to minimize pressure drop.

(83) Additional dimples **122** are located in between dimples **110** and elongate bridge dimples **120**, alongside such side elongate bridge dimples **120**. In addition, additional dimples **122** are located in between dimples **110** and elongate bridge dimples **121**. Dimples **122** maintain clearance between the interconnect **20** and the metal substrate **65** avoiding fuel blockage in such areas where the additional dimples **122** are located.

(84) Elongate bridge dimples **120c** comprise a wedge shape along its shortest side of its rectangular cross-sectional shape, such wedge shape located on the side connecting the fuel distribution channel region **44c3** to the elongate bridge dimples **120c**. In a fuel cell stack assembly, the wedge shape maintains clearance between the metal interconnect plate **20** and the next fuel cell unit **1**, reducing the risk of short circuits between two adjacent solid oxide fuel cell units **1**.

(85) FIGS. **16A**, **16B** and **16C** show the significant improvement in fuel velocity achieved with the

present invention (FIG. 16C) as compared to prior art devices. In particular, there is a significant increase in fuel velocity around port areas. The fuel velocity remains more constant with the present invention (FIG. 16C), with less fuel deficiency regions at the corner of the fuel cell unit 1, and the fuel promoted uniformly across the fuel cell unit 1, which improve the chemical reaction occurring within the fuel cell.

(86) FIGS. 17A and 17B and Table 1 show that in the present invention the normalised fuel residence time has decreased as compared to the prior art, which means less concentration of Hydrogen is required for the chemical reaction. That is to say, less fuel is needed for the chemical reaction to occur, so the efficiency of the present invention is improved as compared to the prior art.

(87) TABLE-US-00001 TABLE 1 Average at Active Average in Maximum region exit active region
Prior art 1.1 0.88 0.72 Embodiment 1 0.78 0.62 0.29

(88) Table 2 illustrates flow uniformity at the fuel cell unit 1 active area—a higher coefficient shows a better flow distribution across the active region, so the fuel is better distributed across the cell in the present application.

(89) TABLE-US-00002 TABLE 2 Entry Exit Prior art 0.56 0.51 Present invention 0.81 0.81

(90) Table 3 illustrates pressure drop at operating point. Due to a better distribution of the fuel, the pressure drop between inlet and outlet ports has decreased in the present invention as compared to the prior art. Minimizing the pressure drop across the cell is beneficial to maintain the compression along the stack.

(91) TABLE-US-00003 TABLE 3 Pressure drop Prior art 58.17 mbar Present invention 49.8 mbar

(92) Suitable material for various components include (Table 4):

(93) TABLE-US-00004 TABLE 4 metal interconnect plate 20 ferritic stainless steel, grade 441 metal spacer 30 ferritic stainless steel, grade 441 blanking plates 50a, 50b Crofer 22 APU (VDM Metals GmbH) metal substrate plates 70a, 70b Crofer 22 APU (VDM Metals GmbH)

EMBODIMENT 2

(94) As shown in FIG. 10, Embodiment 2 is as per Embodiment 1, except that in metal supported solid oxide fuel cell unit 1: (i) blanking plate 50a and metal substrate plate 70a are formed as a combined metal substrate plate 170a, and (ii) blanking plate 50b and metal substrate plate 70b are formed as a combined metal substrate plate 170b.

(95) Fabrication and operation is otherwise identical to that of Embodiment 1.

EMBODIMENT 3

(96) As per WO2015/136295, a fuel cell stack assembly is formed using a plurality of fuel cell units 1. In more detail, a stack of fuel cell units 1 is assembled on top of a metal base plate (ferritic stainless steel 3CR12), with a Thermiculite 866 gasket electrically insulating the base plate from the adjacent fuel cell unit 1, and a power take off located between the Thermiculite 866 gasket and the adjacent fuel cell unit 1. Thermiculite 866 gaskets are located between the first ends 2 of adjacent fuel cell units 1, and between the second end 3 of adjacent fuel cell units. A power take-off is then positioned upon the top (i.e. the exposed) fuel cell unit 1, a Thermiculite 866 gasket is then placed on top of the power take-off and a metal end plate (ferritic stainless steel 3CR12) placed upon the Thermiculite gasket. Compressive force is then exerted by compression means between the base plate and the end plate, and a skirt attached to the base plate and the end plate to define a volume between them within which is contained the fuel cell stack and its fuel cell units.

EMBODIMENT 4

(97) As shown in FIG. 11, Embodiment 4 is as per Embodiment 1, except that in solid oxide fuel cell unit 1: (i) blanking plate 50a, metal substrate plate 70a, metal substrate plate 70b and blanking plate 50b are formed as a single combined metal substrate plate 180 (a metal substrate) (ii) metal spacer 30 has a single cut-out internal perimeter 39a defining a single cut-out.

(98) Fabrication and operation is otherwise identical to that of Embodiment 1.

EMBODIMENT 5

(99) As shown in FIG. 12, a fuel cell unit **1** is fabricated as per Embodiment 1. In this embodiment, there are a total of six metal substrate plates **70**, and six corresponding cut-outs **40**.

EMBODIMENT 6

(100) This embodiment is as per Embodiment 4, except that metal substrate plate **180** comprises a single porous region **78**, and a single fuel cell **79** is provided on the second surface **72** of metal substrate plate **180**, the porous region and the fuel cell extending to the perimeter of the single cut-out internal perimeter **39a**.

(101) When manufacturing the fuel cell unit **1**, the first welding step (in which the metal substrate **65/70** components are welded to the metal spacer **30**) is not necessary. Instead, a single welding around the perimeter through the three layers is performed.

(102) Various modifications, adaptations and alternative embodiments will be readily apparent to the person of ordinary skill in the art without departing from the scope of the appended claims. Reference signs are incorporated in the claims solely to ease their understanding, and do not limit the scope of the claims.

REFERENCE SIGNS

(103) **1** Solid oxide fuel cell unit **2** First end **3** Second end **20** Metal interconnect plate **21** First surface (of metal interconnect plate **20**) **22** Second surface (of metal interconnect plate **20**) **23** External perimeter (of metal interconnect plate **20**) **24** Fuel port (of metal interconnect plate **20**) **27** First edge (of metal interconnect plate **20**) **28** Second edge (of metal interconnect plate **20**) **30** Metal spacer **31** First surface (of metal spacer **30**) **32** Second surface (of metal spacer **30**) **33** External perimeter (of metal spacer **30**) **33a** Fuel inlet internal perimeter **33b** Fuel inlet internal perimeter **33c** Fuel outlet internal perimeter **33d** Fuel outlet internal perimeter **34a** Fuel port **34b** Fuel port **34c** Fuel port **34d** Fuel port **35a** Fuel inlet port volume **35b** Cut-out volume **35c** Fuel outlet port volume **37** First elongate edge (of metal spacer **30**) **38** Second elongate edge (of metal spacer **30**) **39a** Cut-out internal perimeter **39b** Cut-out internal perimeter **40** Cut-out **40a** Cut-out **40b** Cut-out **41** Cross member **44a** Fuel duct region **44a1** Fuel duct region **44a2** Fuel duct region **44a3** Fuel duct region **44b** Fuel throat region **44c** Fuel distributor channel region **44c1** Fuel distributor channel region **44c2** Fuel distributor channel region **44c3** Fuel distributor channel region **50a** Blanking plate **50b** Blanking plate **51** First surface (of blanking plate) **52** Second surface (of blanking plate) **54** Fuel port (of blanking plate) **55** Curved edge (of blanking plate) **57** First edge (of blanking plate) **58** Second edge (of blanking plate) **59** Inner edge (of blanking plate) **65** Metal substrate **65a** Metal substrate first surface **65b** Metal substrate second surface **70** Metal substrate **70a** Metal substrate plate **70b** Metal substrate plate **71** First surface (of metal substrate plate) **72** Second surface (of metal substrate plate) **74** First short side (of metal substrate plate) **75** Second short side (of metal substrate plate) **76** Outer long side (of metal substrate plate) **77** Inner long side (of metal substrate plate) **78** Porous region (of metal substrate plate) **78a** Perforation **78b** Non-porous region (of metal substrate plate) **79** Solid oxide fuel cell **80** Assembly baseplate **81** Datum edge **82a** Tolerance gap **82b** Tolerance gap **82c** Tolerance gap **83a** Fixed dowel **83b** Fixed dowel **83c** Fixed dowel **83d** Fixed dowel **83e** Fixed dowel **83f** Fixed dowel **83g** fixed dowel **84a** Spring loaded dowel **84b** Spring loaded dowel **84c** Spring loaded dowel **84d** Spring loaded dowel **84e** Spring loaded dowel **84f** Spring loaded dowel **84g** Spring loaded dowel **84h** Spring loaded dowel **90** First clamping plate **91a** Welding slot **91b** Welding slot **91c** Welding slot **92a** Orifice **92b** Orifice **95** Second clamping plate **96** Inner perimeter **96a** Opening **98a** Orifice **98b** Orifice **100a** Line weld seam **100b** Line weld seam **100c** Line weld seam **100d** Line weld seam **100e** Line weld seam **100f** Line weld seam **101** Perimeter weld seam **110** Dimple **120** Elongate bridge dimple **120a** Elongate bridge dimple **120b** Elongate bridge dimple **120c** Elongate bridge dimple **121** Elongate bridge dimple **122** Dimple **130** Fuel duct **140** Fluid flow path **170a** Combined metal substrate plate **170b** Combined metal substrate plate **180** Combined metal substrate plate **da** Distance **db** Distance **dc** Distance **rc1** Width **rc2** Width **rc3** Width **L** Length (of fuel throat region **44b**) **W** Width (of fuel throat region **44b**)

Claims

1. A metal supported solid oxide fuel cell unit comprising: a) a metal substrate defining first and second opposed surfaces, wherein at least one solid oxide fuel cell is disposed on said second surface of said metal substrate; b) a metal spacer, which defines first and second opposed surfaces, said metal spacer comprising: (i) an external perimeter, (ii) at least one fuel inlet internal perimeter defining a fuel inlet port, (iii) at least one cut-out internal perimeter defining a cut-out, and (iv) at least one fuel outlet internal perimeter defining a fuel outlet port, wherein said first surface of said metal substrate is attached to said second surface of said metal spacer; and c) a metal interconnect plate which defines first and second opposed surfaces, said second surface of said metal interconnect plate sealingly attached to said first surface of said metal spacer, wherein: a fuel inlet port volume is defined between said first surface of said metal substrate, each fuel inlet internal perimeter of said metal spacer, and said second surface of said metal interconnect plate, a cut-out volume is defined between said first surface of said metal substrate, said at least one cut-out internal perimeter of said metal spacer, and said second surface of said metal interconnect plate, and a fuel outlet port volume is defined between said first surface of said metal substrate, each fuel outlet internal perimeter of said metal spacer, and said second surface of said metal interconnect plate, wherein said metal interconnect plate comprises a plurality of bridge portions defining a fluid flow path from said at least one fuel inlet port volume to said at least one cut-out volume to said at least one fuel outlet port volume.
2. A metal supported solid oxide fuel cell unit according to claim 1, wherein a fluid flow path is defined from the at least one fuel inlet port to the at least one cut-out internal perimeter to the at least one fuel outlet port via the bridge portions.
3. A metal supported solid oxide fuel cell unit according to claim 1, wherein there are a plurality of bridge portions between adjacent volumes.
4. A metal supported solid oxide fuel cell unit according to claim 1, wherein each metal spacer fuel inlet port and each metal spacer fuel outlet port comprises a fuel duct region, a plurality of fuel throat regions, and a corresponding plurality of fuel distributor channel regions.
5. A metal supported solid oxide fuel cell unit according to claim 1, wherein said bridge portions extend outwardly from said first surface of said metal interconnect plate, away from said second surface of said metal interconnect plate.
6. A metal supported solid oxide fuel cell unit according to claim 1, wherein said bridge portions comprise an elongate dimple.
7. A metal supported solid oxide fuel cell unit according to claim 1, wherein said bridge portions define a volume between said first surface of said metal spacer and said second surface of said metal interconnect plate.
8. A metal supported solid oxide fuel cell unit according to claim 1, wherein said metal spacer comprises at least two fuel inlet internal perimeters defining at least two fuel inlet ports.
9. A metal supported solid oxide fuel cell unit according to claim 1, wherein said metal spacer comprises at least two cut-out internal perimeters defining at least two cut-outs.
10. A metal supported solid oxide fuel cell unit according to claim 1, wherein said metal spacer comprises at least two fuel outlet internal perimeters defining at least two fuel outlet ports.
11. A metal supported solid oxide fuel cell unit according to claim 1, wherein said metal supported solid oxide fuel cell unit is a metal supported solid oxide fuel cell stack layer.
12. A solid oxide fuel cell stack comprising a plurality of metal supported solid oxide fuel cell units according to claim 1.
13. A solid oxide fuel cell stack assembly comprising: a base plate, an end plate, a solid oxide fuel cell stack according to claim 12, and a skirt attached to said base plate and said end plate and defining a volume between said skirt, said base plate and said end plate within which is contained

said fuel cell stack.

14. A method of assembly of a metal supported solid oxide fuel cell unit, the metal supported solid oxide fuel cell unit comprising: a) a metal substrate defining first and second opposed surfaces, wherein at least one solid oxide fuel cell is disposed on said second surface of said metal substrate; b) a metal spacer, which defines first and second opposed surfaces, said metal spacer comprising: (i) an external perimeter, (ii) at least one fuel inlet internal perimeter defining a fuel inlet port, (iii) at least one cut-out internal perimeter defining a cut-out, and (iv) at least one fuel outlet internal perimeter defining a fuel outlet port; and c) a metal interconnect plate which defines first and second opposed surfaces, said method of assembly comprising the steps of: (i) attaching said first surface of said metal substrate to said second surface of said metal spacer; and (ii) sealingly attaching said second surface of said metal interconnect plate to said first surface of said metal spacer, wherein: a fuel inlet port volume is defined between said first surface of said metal substrate, each fuel inlet internal perimeter of said metal spacer, and said second surface of said metal interconnect plate, a cut-out volume is defined between said first surface of said metal substrate, said at least one cut-out internal perimeter of said metal spacer, and said second surface of said metal interconnect plate, and a fuel outlet port volume is defined between said first surface of said metal substrate, each fuel outlet internal perimeter of said metal spacer, and said second surface of said metal interconnect plate, wherein said metal interconnect plate comprises a plurality of bridge portions defining a fluid flow path from said at least one fuel inlet port volume to said at least one cut-out volume to said at least one fuel outlet port volume.

.Iadd.15. A metal supported solid oxide fuel cell unit, comprising: a) a metal substrate defining first and second opposed surfaces, wherein at least one solid oxide fuel cell is disposed on said second surface of said metal substrate; b) a metal spacer, which defines first and second opposed surfaces, said metal spacer comprising a port internal perimeter defining a port, wherein said first surface of said metal substrate is attached to said second surface of said metal spacer; and c) a metal interconnect plate which defines first and second opposed surfaces, said second surface of said metal interconnect plate sealingly attached to said first surface of said metal spacer; wherein: a port volume is defined between said first surface of said metal substrate, said port internal perimeter of said metal spacer, and said second surface of said metal interconnect plate; said metal interconnect plate comprises a bridge portion defining a fluid flow path through which said port volume is coupled to an adjacent volume in the solid oxide fuel cell unit; and said bridge portion defines a volume between said first surface of said metal spacer and said second surface of said metal interconnect plate. .Iaddend.

.Iadd.16. A metal supported solid oxide fuel cell unit according to claim 15, wherein the adjacent volume is an active region of the solid oxide fuel cell unit. .Iaddend.

.Iadd.17. A metal supported solid oxide fuel cell unit according to claim 15, wherein there are a plurality of bridge portions through which said port volume is coupled to said adjacent volume. .Iaddend.

.Iadd.18. A metal supported solid oxide fuel cell unit according to claim 15, wherein said port volume comprises a duct region, a plurality of throat regions, and a corresponding plurality of distributor channel regions. .Iaddend.

.Iadd.19. A metal supported solid oxide fuel cell unit according to claim 18, wherein each throat region has a constant width between the duct region and the corresponding distributor channel region. .Iaddend.

.Iadd.20. A metal supported solid oxide fuel cell unit according to claim 15, wherein said bridge portion extends outwardly from said first surface of said metal interconnect plate, away from said second surface of said metal interconnect plate. .Iaddend.

.Iadd.21. A metal supported solid oxide fuel cell unit according to claim 15, wherein said bridge portion comprises an elongate dimple. .Iaddend.

.Iadd.22. A metal supported solid oxide fuel cell unit according to claim 15, wherein the port is an

inlet port. .Iaddend.

.Iadd.23. A metal supported solid oxide fuel cell unit according to claim 15, wherein the port is an outlet port. .Iaddend.
